DATA SCIENCE & ARTIFICIAL INTELLIGENCE



Probability and Statistics
Statistics - II

Lecture No. 07



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Recap of previous lecture



Topic

1. SAMPLING
2. Z-Distribution
3. Z-test

Topics to be Covered



Topic

t-Distributions & t-test

(RECAP)

i. Sample proportion $(\tilde{p}) = \frac{x}{n} = \frac{\text{No. of success in sample}}{\text{sample size}}$ (x = no. of success) & $\tilde{q} = 1 - \tilde{p}$

& Population Proportion $(p_0) = \frac{X}{N} = \frac{\text{No. of success in population}}{\text{Population size}}$

⊛ Sample Mean $(\bar{x})$ & Sample proportion $(\tilde{p})$ are Normally Distributed by C.L.Th

Standard Error → it is the S.D of Sample Mean & Sample Proportion

① SE of Mean $= SE(\bar{x}) = \frac{\sigma}{\sqrt{n}}$

② SE of Proportion $= SE(\tilde{p}) = \sqrt{\frac{\tilde{p}\tilde{q}}{n}}$

Most Probable limits → $\bar{x} - 3SE(\bar{x}) \le \mu \le \bar{x} + 3SE(\bar{x})$ & $\tilde{p} - 3SE(\tilde{p}) \le p_0 \le \tilde{p} + 3SE(\tilde{p})$

① P.T. $SE(\bar{x}) = \frac{\sigma}{\sqrt{n}}$

Proof: w.k.that,

$$\bar{x} = \frac{x_1+x_2+x_3+\cdots+x_n}{n}$$

$$Var(\bar{x}) = Var\left\{\frac{x_1+x_2+x_3+\cdots+x_n}{n}\right\}$$

$$= \frac{1}{n^2} Var\{x_1+x_2+x_3+\cdots+x_n\}$$

$$= \frac{1}{n^2}\{Var(x_1)+Var(x_2)+\cdots+Var(x_n)\}$$

$$= \frac{1}{n^2}\{\sigma^2+\sigma^2+\sigma^2+\cdots+\sigma^2\}$$

$$= \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n} \Rightarrow SE(\bar{x}) = \frac{\sigma}{\sqrt{n}}$$

② P.T $SE(\bar{p}) = \sqrt{\frac{\bar{p}\bar{q}}{n}}$

Proof: $Var(\bar{p}) = Var\left(\frac{x}{n}\right)$

$$= \frac{1}{n^2} Var(x)$$

$$= \frac{1}{n^2}[n\bar{p}\bar{q}] = \frac{\bar{p}\bar{q}}{n}$$

$$\therefore SE(\bar{p}) = \sqrt{\frac{\bar{p}\cdot\bar{q}}{n}}$$

#Q. In previous question, if success is occurring 3240 times then Prove that die is biased.

HW8

$x = \{$ No. of times 1 or 6 is obtained $\}$ $\begin{cases} x = 3240 \\ n = 9000 \end{cases}$ $\bar{p} = \frac{x}{n} = \frac{3240}{9000} = 0.36$

$\bar{q} = 1 - \bar{p} = 0.64$

$\therefore SE(\bar{p}) = \sqrt{\frac{\bar{p}\bar{q}}{n}} = \sqrt{\frac{0.34 \times 0.66}{9000}} = 0.005$, i.e. $3SE(\bar{p}) = 0.015$

Most probable limits of p_0 are $\bar{p} - 3SE(\bar{p}) \le p_0 \le \bar{p} + 3SE(\bar{p})$

$0.360 - 0.015 \le p_0 \le 0.360 + 0.015$

$0.345 \le p_0 \le 0.375$

But for unbiased die, $p_0 = P(1 \text{ or } 6) = \frac{2}{6} = \frac{1}{3} = 0.333$

$\therefore p_0$ is not with in the Probable limits $\therefore$ Die is BIASED.

(RECAP)

Generally, it is represented in terms of % and it is predecided in Question

Confidence Region	Z-Statistic
68.28%	$\|Z\| \le 1 \Rightarrow -1 \le Z \le 1$
80%	$\|Z\| \le 1.28$
90%	$\|Z\| \le 1.645$
✓ 95%	$\|Z\| \le 1.96$
95.5%	$\|Z\| \le 2 \Rightarrow -2 \le Z \le 2$
98%	$\|Z\| \le 2.33$
✓ 99%	$\|Z\| \le 2.58$
99.7%	$\|Z\| \le 3 \Rightarrow -3 \le Z \le 3$

(RECAP)



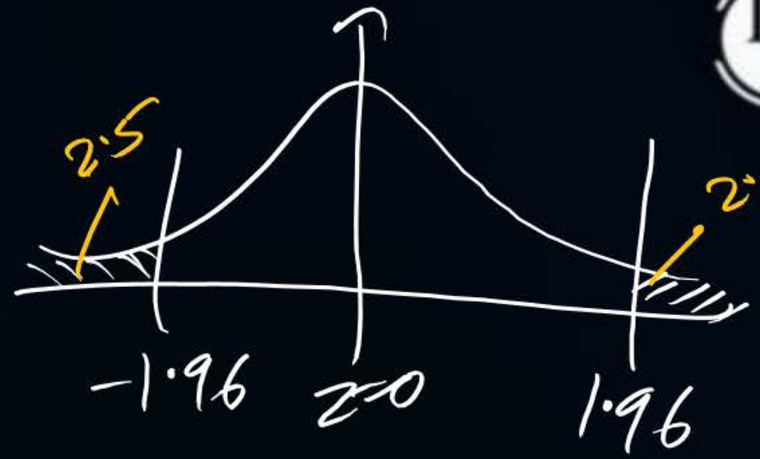
$\alpha = 0.1$

& $Z_\alpha(0.1) = 1.28$



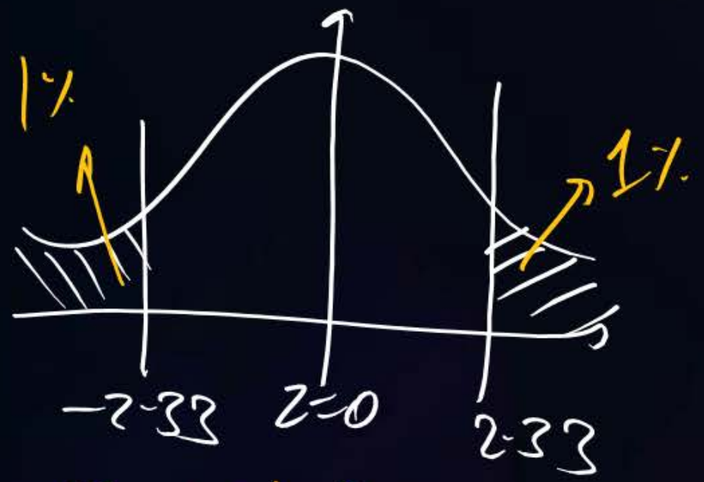
$\alpha = 0.05$

$Z_\alpha(0.05) = 1.645$

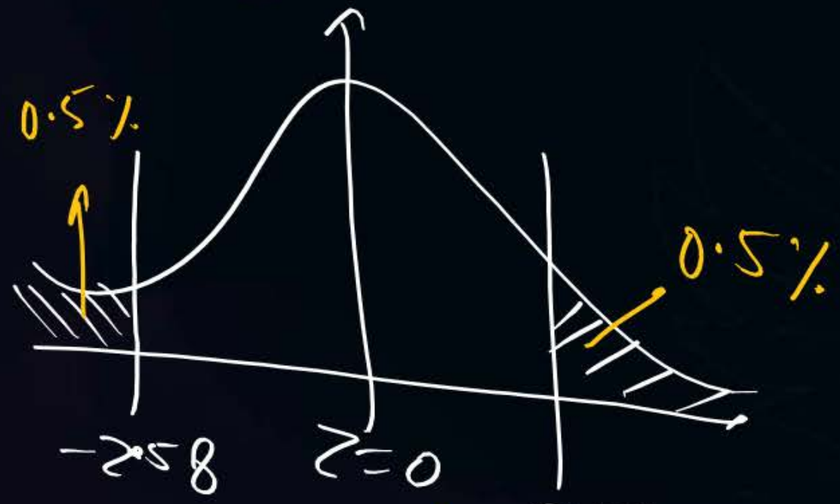


$\alpha = 0.025$

$Z_\alpha(0.025) = 1.96$



$\alpha = 0.01$ & $Z_\alpha(0.01) = 2.33$



$\alpha = 0.005$ & $Z_\alpha(0.005) = 2.58$

(RECAP)

with the help of Z-test, we can solve following Types of Questions:

Type 1 Significance of Population Mean

$H_0 : \mu = \mu_0$ & $H_1 : \mu \neq \mu_0$

↳ ie there is No significant difference b/n Sample mean & Population Mean.

$$Z = \frac{\bar{x} - \mu_0}{SE(\bar{x})} = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

If Z lies in the Confidence Region, we will accept H_0 otherwise we Reject H_0

Type 2 Significance of Difference b/n Two Population Mean →

$H_0 : \mu_1 = \mu_2$ & $H_1 : \mu_1 \neq \mu_2$

$$Z = \frac{\bar{x} - \bar{y}}{SE(\bar{x} - \bar{y})} = \frac{\bar{x} - \bar{y}}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

(RECAP)

Type 3 : Significance of population proportion

$H_0 : \tilde{p} = p_0$ & $H_1 : \tilde{p} \neq p_0$

↳ ie there is no significant diff b/n Sample prop ($\tilde{p}$) & pop. prop (p_0)

$$Z = \frac{x-\mu}{\sigma} = \frac{x-(np_0)}{\sqrt{np_0q_0}} = \frac{n\left(\frac{x}{n}-p_0\right)}{\sqrt{np_0q_0}}$$

$$= \frac{\left(\frac{x}{n}\right)-p_0}{\frac{1}{n}\sqrt{np_0q_0}} = \frac{\tilde{p}-p_0}{\sqrt{\frac{p_0q_0}{n}}}$$

Type 4 Significance of difference b/n two population proportion

$H_0 : p_1 = p_2$ & $H_1 : p_1 \neq p_2$

↳ ie there is no significant difference b/n two population proportion

$$\tilde{p}_1 = \frac{x_1}{n_1}, \quad \tilde{p}_2 = \frac{x_2}{n_2}, \quad \tilde{p} = \frac{x_1+x_2}{n_1+n_2}$$

where $\tilde{p}$ = common proportion of two samples

$$Z = \frac{\tilde{p}_1-\tilde{p}_2}{SE(\tilde{p}_1-\tilde{p}_2)} = \frac{\tilde{p}_1-\tilde{p}_2}{\sqrt{\frac{\tilde{p}\tilde{q}}{n_1}+\frac{\tilde{p}\tilde{q}}{n_2}}}$$

#Q. In a town, 350 out of 600 person were found to be vegetarian. On the basis of this data, can we say that majority of population in the town in vegetarian at 5% level of significance

HW

Given that for right tailed $Z_\alpha(0.05) = 1.645$



$x = \{$ No. of Vegie persons $\}$ $n = 350$, $n = 600$

$\therefore \tilde{p} = \frac{x}{n} = \frac{350}{600} = 0.58$

← By Common sense $p_0 = 0.5$, $q_0 = 0.5$

H_0 : No. of Vegie persons = No. of Non Vegie Persons

ie $H_0 : \tilde{p} = p_0$

H_1 : No. of Vegie persons > No of Non Vegie persons

ie $H_0 : \tilde{p} > p_0$ (ie it is Right Tailed)

$$Z = \frac{\tilde{p} - p_0}{\sqrt{\frac{p_0 q_0}{n}}} = \frac{0.58 - 0.50}{\sqrt{\frac{0.5 \times 0.5}{600}}} = 3.2$$

ie Reject H_0 & Accept H_1 ie Ans YES

#Q. A machine produced 20 defective articles in a lot of 400 and after overhauling it produced 20 defective articles in a lot of 300. Has the machine improved at $\alpha = 0.01$.

HW

For right tailed $Z_\alpha(0.01) = 2.33$

$x = \{$ No. of defective Articles in a lot $\}$ → success.

$x_1 = 20, n_1 = 400, \bar{p}_1 = \frac{x_1}{n_1} = \frac{20}{400} = 0.050$

$x_2 = 20, n_2 = 300, \bar{p}_2 = \frac{x_2}{n_2} = \frac{20}{300} = 0.066$

Common proportion is $\bar{p} = \frac{x_1 + x_2}{n_1 + n_2}$

ie $\bar{p} = \frac{20+20}{400+300} = \frac{40}{700} = 0.057$

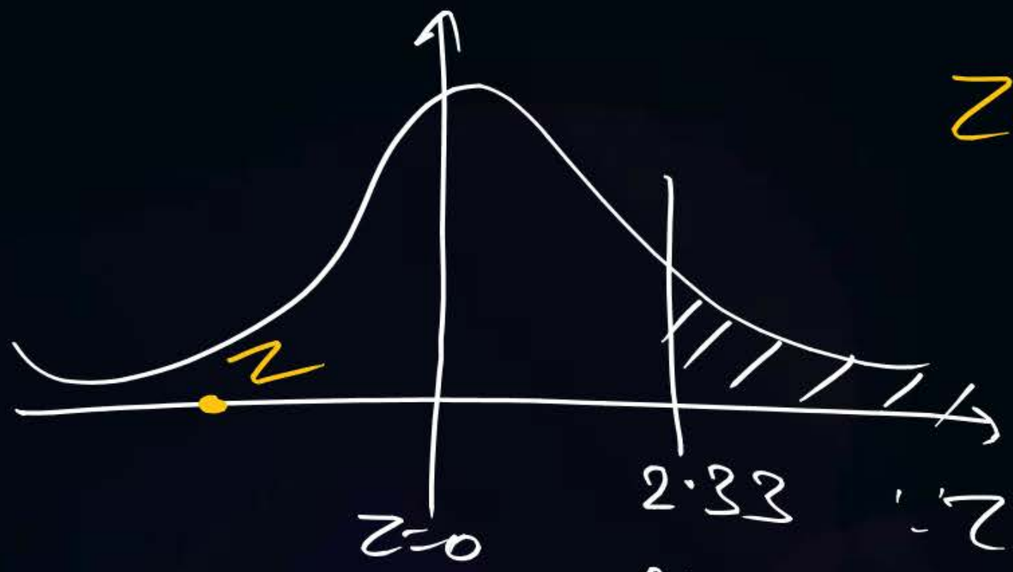
& $\bar{q} = 0.943$

p_1 = Population Proportion of Def. Articles before overhauling

p_2 = " " " " " after " "

$H_0 : \boxed{p_1 = p_2}$ i.e Machine has not been improved

$H_1 : \boxed{p_1 > p_2}$ i.e Machine has been improved



$$Z = \frac{\tilde{p}_1 - \tilde{p}_2}{\sqrt{\frac{\tilde{p}\tilde{q}}{n_1} + \frac{\tilde{p}\tilde{q}}{n_2}}} = \frac{0.050 - 0.066}{\sqrt{0.054 \times 0.943\left(\frac{1}{400} + \frac{1}{300}\right)}}$$

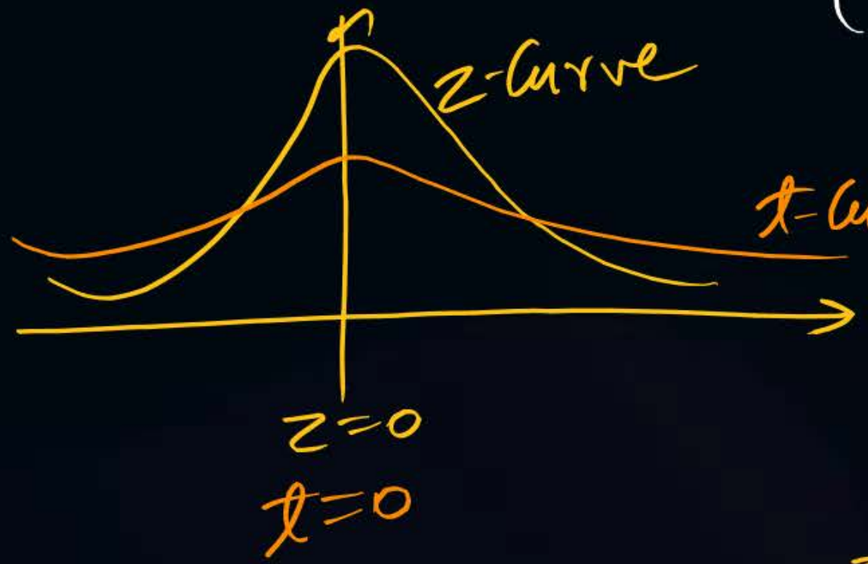
$$= -0.94$$

$\therefore$ Z lies in Acceptance Region $\therefore$ H_0 is Accepted

i.e Machine has not been improved.

t-Distribution & Student's t-test

(Small Sample test is for $n < 30$)



$$\text{Population Mean} = \frac{\sum X}{N} = \mu$$

$$\text{Sample Mean} = \frac{\sum x}{n} = \bar{x}$$

$$\text{Population Variance} = \frac{\sum (X_i - \mu)^2}{N} = \sigma^2$$

while

$$\text{Sample Variance} = \frac{\sum (x_i - \bar{x})^2}{(n-1)} = s^2$$

(*) Reason why we divide with (n-1) while Calculating Sample Variance →

In Case of Sample Distribution mean is assumed as one of the Sample point, so we have only (n-1) differences in Numerator, Hence we should divide with (n-1).

Mathematically it can be understand by analysing following two Cases;

Case I: $S^2 = \frac{\sum(x_i-\bar{x})^2}{n} = \frac{(x_1-\bar{x})^2+(x_2-\bar{x})^2+\ldots+(\bar{x}-\bar{x})^2+\ldots+(x_n-\bar{x})^2}{n}$

$= \frac{\sigma^2+\sigma^2+\sigma^2+\ldots+0^2+\ldots+\sigma^2}{n} = \frac{(n-1)\sigma^2}{n} \neq \sigma^2$ ☹

Case II: $S^2 = \frac{\sum(x_i-\bar{x})^2}{(n-1)} = \frac{(x_1-\bar{x})^2+(x_2-\bar{x})^2+\ldots+(\bar{x}-\bar{x})^2+\ldots+(x_n-\bar{x})^2}{(n-1)}$

$= \frac{\sigma^2+\sigma^2+\sigma^2+\ldots+0^2+\ldots+\sigma^2}{(n-1)} = \frac{(n-1)\sigma^2}{(n-1)} = \sigma^2$ ☺

Degree of Freedom → (ν) = Total No. of observations − No. of conditions imposed on them

i.e DF = "Max Number of Independent Variables that have freedom to Vary"

for eg ① if we are taking 5 Points for which Sum = 40 then we have liberty to Choose only 4 points as 5th Point = 40 − (Sum of 4 Points)

∴ In this situation, Df (ν) = 5 − 1 = 4

for eg ② If we are taking n Boys for which their Average wt = 65 kg & S.D (σ) = 9 kg then Df (ν) = n − 2

Note ① for one Sample t-test, Df = n − 1 & for two Sample t-test Df = n − 2

② for large Df, t-Distribution Converts into Normal Distribution.

with the help of t-test we can deal following types of problems →

Type 1 Significance of population Mean:

$H_0 : \mu = \mu_0$; $H_1 : \mu \neq \mu_0$

$$t = \frac{\bar{x} - \mu_0}{SE(\bar{x})} = \frac{\bar{x} - \mu_0}{\sqrt{s^2/n}} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

where $Df = (n-1)$

& $s^2 = \frac{\Sigma(x-\bar{x})^2}{n-1}$

& $SE(\bar{x}) = s/\sqrt{n}$

Type 2 Significance of difference b/w two pop mean →

$H_0 : \mu_1 = \mu_2$, $H_1 : \mu_1 \neq \mu_2$

$s_1^2 = \frac{\Sigma(x-\bar{x})^2}{n_1-1}$ & $s_2 = \frac{\Sigma(y-\bar{y})^2}{n_2-1}$

Common Sample Variance is

$$s^2 = \frac{\Sigma(x-\bar{x})^2 + \Sigma(y-\bar{y})^2}{(n_1+n_2)-2} = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}$$

$$t = \frac{\bar{x} - \bar{y}}{\sqrt{\frac{s^2}{n_1} + \frac{s^2}{n_2}}}$$

where $d.f = (n_1+n_2-2)$

#Q. A random sample of 16 values from normal population has mean of 41.5 cm and sum of squares of deviation from mean is 135 cm^2. ~~can we say that the popular mean is 43.5 cm?~~ can we say that the population mean is 43.5 cm? with 5% level of significance.

(ii) Also find the 95% and 99% confidence for μ it is given that t_{15} (0.05) = 2.131 and $t_{15}(0.01) = 2.94$

Type 1

$n=16$, $\bar{x}=41.5$, $\sum(x-\bar{x})^2=135$, $\mu_0=?$

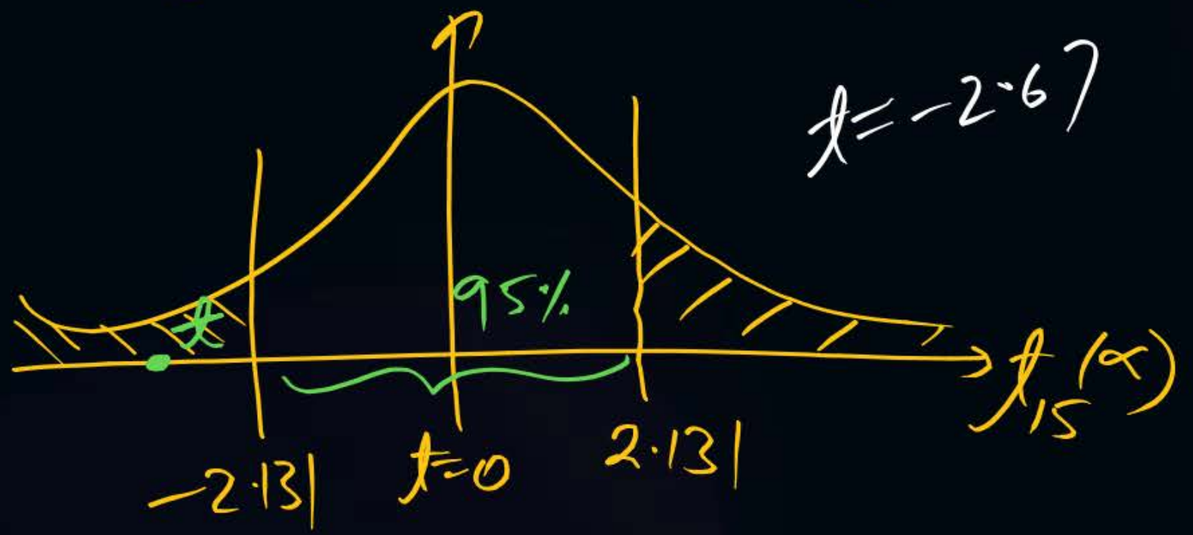
$Df(\nu)=n-1=15$, $s^2=\frac{\sum(x-\bar{x})^2}{n-1}=\frac{135}{16-1}=9 \Rightarrow s=3$

$H_0: \mu_0=43.5$

$H_1: \mu_0 \neq 43.5$

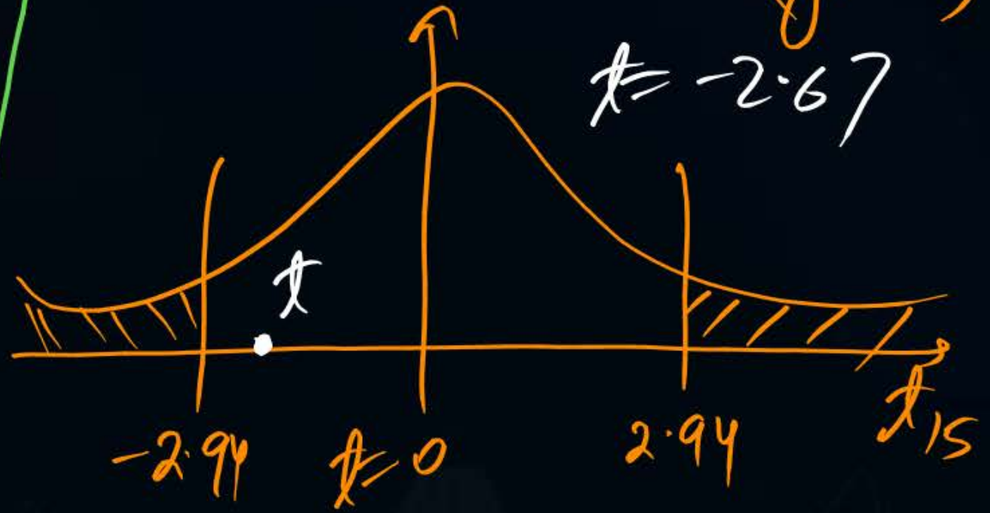
Now $t=\frac{\bar{x}-\mu_0}{s/\sqrt{n}}=\frac{41.5-43.5}{3/\sqrt{16}}=\frac{2\times4}{3}=\frac{8}{3}=2.67$

(i) $t_{15}(0.05) = 2.131$ (given)



$\therefore$ t lies in Rejection Region so
Reject H_0 & Accept H_1
i.e $\mu \neq 43.5$ for 5% level of significance

(ii) $t_{15}(0.01) = 2.94$ (given)



$\therefore$ t lies in Confidence Region so
Accept H_0 & Reject H_1 i.e
$\mu = 43.5$ for 1% level of
significance

(iii) Also find the 95% & 99% confidence limits for μ ?

Sol: for 95%: $t_{15}(0.05) = 2.131$

$$SE(\bar{x}) = \frac{\sigma}{\sqrt{n}} = \frac{3}{\sqrt{16}} = \frac{3}{4} = 0.75$$

So Most probable limits for μ are

$$\bar{x} - t_{15}(0.05).SE(\bar{x}) \le \mu \le \bar{x} + t_{15}(0.05).SE(\bar{x})$$

$$41.5 - 2.131(0.75) \le \mu \le 41.5 + 2.131(0.75)$$

$$\boxed{39.90 \le \mu \le 43.09}$$

for 99%, $t_{15}(0.01) = 2.94$

Hence Most probable limits for μ are

$$\bar{x} - t_{15}(0.01).SE(\bar{x}) \le \mu \le \bar{x} + t_{15}(0.01).SE(\bar{x})$$

$$41.5 - 2.94(0.75) \le \mu \le 41.5 + 2.94(0.75)$$

$$\boxed{39.29 \le \mu \le 43.71}$$

V.V. Sp. Note: for Z Dist, Find 99.7%, 99% & 95% Confidence limits for μ

(i) $\bar{x} - 3\,SE(\bar{x}) \le \mu < \bar{x} + 3\,SE(\bar{x})$ $\because Z(0.003) = 3$

(ii) $\bar{x} - 2.58\,SE(\bar{x}) \le \mu \le \bar{x} + 2.58\,SE(\bar{x})$ $\because Z(0.01) = 2.58$

(iii) $\bar{x} - 2.33\,SE(\bar{x}) \le \mu \le \bar{x} + 2.33\,SE(\bar{x})$ $\because Z(0.05) = 2.33$

Similarly in t-Dist, Find 99% & 95% Confidence limit for Df = 15

(i) $\bar{x} - 2.94\,SE(\bar{x}) \le \mu \le \bar{x} + 2.94\,SE(\bar{x})$ $\because t_{15}(0.01) = 2.94$

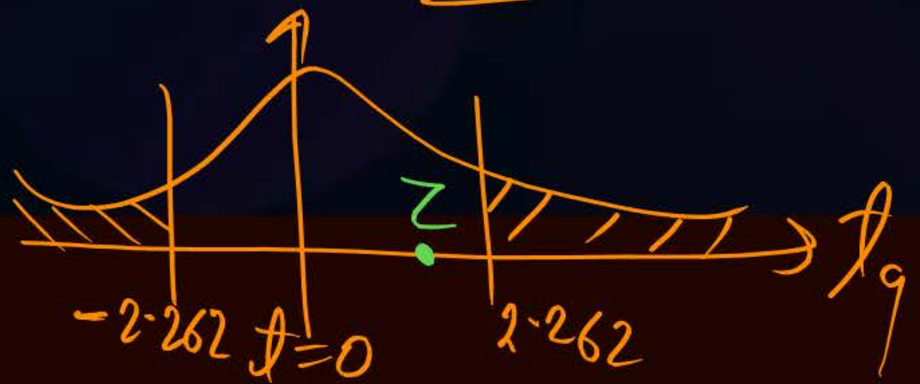
(ii) $\bar{x} - 2.131\,SE(\bar{x}) \le \mu \le \bar{x} + 2.131\,SE(\bar{x})$ $\because t_{15}(0.05) = 2.131$

#Q. Average height of 10 student is a school is observed as 67 inches with sum of the squares of deviations from central value is 88. can we say that average height of student in a school is 65 inches. It is given that, value of t at 5% level of significance with 9 degree of freedom is 2.262.

$n=10,\ \bar{x}=67,\ \sum(x-\bar{x})^2=88,\ \mu=?$

$\nu=9,\ s^2=\frac{\sum(x-\bar{x})}{n-1}=\frac{88}{10-1}=\frac{88}{9}=9.77 \Rightarrow s=\sqrt{9.77}$

$H_0: \mu=65,\ H_1: \mu\neq 65,\ t=\frac{\bar{x}-\mu_0}{s/\sqrt{n}}=\frac{\bar{x}-\mu_0}{\sqrt{s^2/n}}=\frac{67-65}{\sqrt{9.77/10}}=2.02$



∵ t lies in Acceptance Region ∴ Accept H_0 & Reject H_1

i.e Conclusion is: Population Mean $\mu=65$

#Q. A machine produces washers of thickness 10mm. A sample of 10 washers has an average thickness of 9.52 mm with SD of 0.6 mm. Find out 't'.

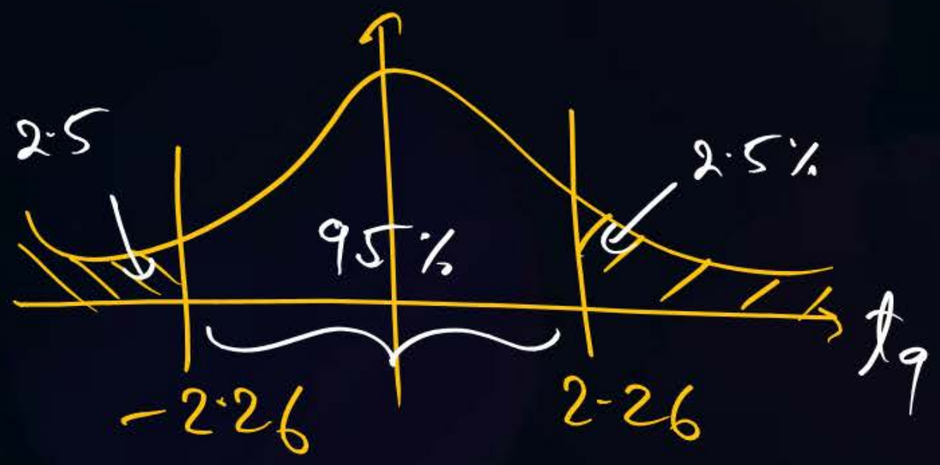
$\mu_0 = 10$, $n = 10$, $\bar{x} = 9.52$, $\beta = 0.6$

$\nu = 9$

$$t = \frac{\bar{x} - \mu_0}{\beta/\sqrt{n}} = \frac{9.52 - 10}{0.6/\sqrt{10}} = -2.51$$

#Q. If the mean and variance of sample of 10 observation are 66 and 14.67 respectively then find the 95% confidence limits for population mean (μ). Given that upper 2.5% point of t distribution with 9 degree of freedom is 2.26

$n = 10 \Rightarrow \nu = n-1 = 9,\ \bar{x} = 66,\ s^2 = 14.67,\ \mu = ?,\ t_9(0.025) = 2.26$



$$SE(\bar{x}) = \frac{s}{\sqrt{n}} = \sqrt{\frac{s^2}{n}} = \sqrt{\frac{14.67}{10}} = 1.21$$

$$\bar{x} - t_9(0.05) \times SE(\bar{x}) \le \mu \le \bar{x} + t_9(0.05) \times SE(\bar{x})$$

$$66 - (2.26)(1.21) \le \mu \le 66 + (2.26)(1.21)$$

$$63.26 \le \mu \le 68.73$$

#Q. The means of two random samples of size 9 and 7 are 196.42 and 198.82 respective variance are 26.94 and 18.73. can we say that the sample are drawn from same normal population?

Type 2

$t_{14}(0.05) = 2.145$

$n_1 = 9, \quad \bar{x}_1 = 196.42, \quad s_1^2 = 26.94$

$n_2 = 7, \quad \bar{x}_2 = 198.82, \quad s_2^2 = 18.73$

$Df(\nu) = n_1 + n_2 - 2 = 9 + 7 - 2 = 14$

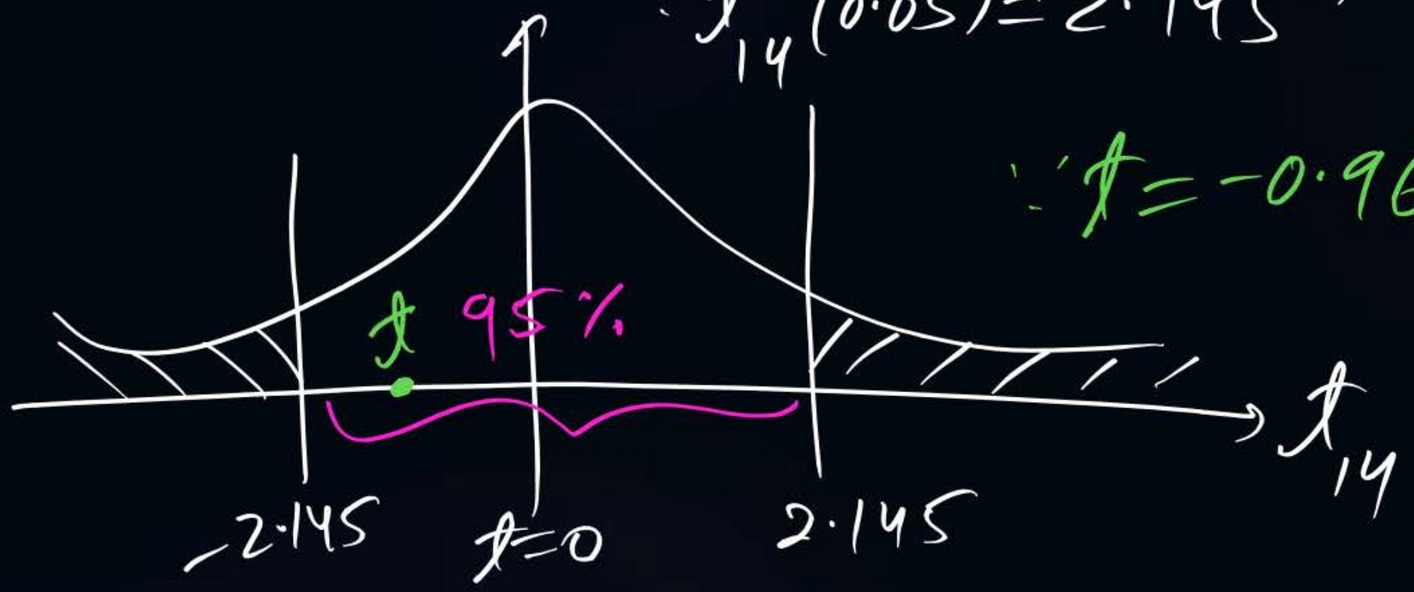
$$s^2 = \frac{s_1^2(n_1-1) + s_2^2(n_2-1)}{n_1+n_2-2} = 23.42$$

$H_0 : \mu_1 = \mu_2$ i.e samples are drawn from same population

$H_1 : \mu_1 \neq \mu_2$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s^2}{n_1} + \frac{s^2}{n_2}}} = \frac{196.42 - 198.82}{23.42\sqrt{\frac{1}{9} + \frac{1}{7}}}$$

$$= -0.96$$



$\because t = -0.96$ i.e. t lies in Confidence Region

So Accept H_0 & Reject H_1

i.e. $\mu_1 = \mu_2 \Rightarrow$ Samples are drawn from same population.

Central Limit Theorem → (C.L.T) →

Consider a population (Not Necessarily Normal) with Mean = μ & Var = σ^2

Now let us consider various samples (each of size $n \geq 30$) as follows;

$S_1 = \{x_1, x_2, x_3, \ldots x_{30}\} \Rightarrow \bar{x}_1$

$S_2 = \{y_1, y_2, y_3 \ldots y_{30}\} \Rightarrow \bar{x}_2$

$S_3 = \{z_1, z_2, z_3 \ldots z_{30}\} \Rightarrow \bar{x}_3$

$\vdots$

$S_{1000} = \{t_1, t_2, t_3 \ldots t_{30}\} \Rightarrow \bar{x}_{1000}$

i.e we are consider a New Distribution formed by sample means as follows;

$\{\bar{x}_1, \bar{x}_2, \bar{x}_3, \bar{x}_4 \ldots\ldots, \bar{x}_{1000}\}$

then this New Distribution formed $\bar{x}_i$ is Normally Distributed with Mean $\simeq \mu$ & Var $= \frac{\sigma^2}{n}$ i.e $\bar{x}_i \sim N\left\{\mu, \frac{\sigma^2}{n}\right\}$

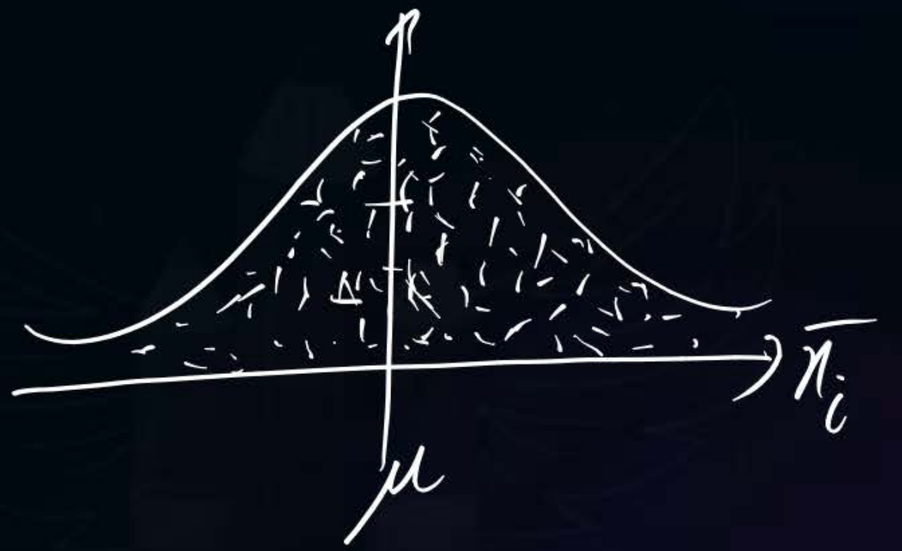
Conclusions: ① Distribution formed by Sample Means is Normally Distributed.

② Average of Distribution formed by Sample Means is almost equal to Population Mean

③ Variance of Distribution formed by Sample Means = σ^2/n

Necessary Conditions for C.L.T →

① Sample Members should be selected at Random
② Sample Members must be Independent
③ Population Members should be distributed identically
④ $30 \leq$ Sample size $< 10\%$ of Population size



Diagram

⊛ C.L.T is applicable for Sample Means & Sample Proportion

Here $\bar{x}_i = \dfrac{\bar{x}_1 + \bar{x}_2 + \bar{x}_3 + \ldots + \bar{x}_{1000}}{n}$

$$E(\bar{x}_i) = E\left\{\frac{\bar{x}_1 + \bar{x}_2 + \bar{x}_3 + \ldots + \bar{x}_{1000}}{n}\right\}$$

$$= \frac{1}{n}\left\{E(\bar{x}_1) + E(\bar{x}_2) + \ldots + E(\bar{x}_{1000})\right\}$$

$$\simeq \frac{1}{n}\left\{\mu + \mu + \mu + \ldots + \mu\right\}$$

$$\simeq \mu$$

ie $\boxed{E(\bar{x}_i) \simeq \mu}$

ie Av of Sample Means $\simeq$ Pop. Mean.

& $Var(\bar{x}_i) = Var\left\{\dfrac{\bar{x}_1 + \bar{x}_2 + \ldots \bar{x}_{1000}}{n}\right\}$

$$= \frac{1}{n^2}\left\{\sigma^2 + \sigma_2^2 + \sigma^2 + \ldots + \sigma^2\right\}$$

$$= \left(\frac{\sigma^2}{n}\right) \text{ ie } \boxed{Var(\bar{x}_i) \simeq \frac{\sigma^2}{n}}$$

Soo $\bar{x}_i$ is also N.R.V

& $\bar{x}_i \sim N\left\{\mu, \frac{\sigma^2}{n}\right\}$

Q. C.L.T does not Hold in which Case :

(a) Sample size ≥ 30 & Population is Normal (T)

(b) " " ≥ 30 & " is not Normal (T)

(c) " " < 30 & population is Normal (T)

(d) ✓ " " < 30 & " is not Normal (f)

Qs A set of samples have been collected from large data and the sample Mean values are 12·8, 10·9, 11·4, 14·2, 12·5, 13·6, 15, 9, 12·6 then find population mean.

a/c. that (By C-L-Th), Average of Sample Means $\equiv$ pop. Mean (μ)

So $$\mu = \frac{\bar{x}_1 + \bar{x}_2 + \bar{x}_3 + \ldots \bar{x}_9}{9}$$

$$= \frac{12.8 + 10.9 + 11.4 + 14.2 + 12.5 + 13.6 + 15 + 9 + 12.6}{9} = 12.4$$

Q In a survey of LKO city, it was Reported that, Average age of Mobile user is of 30 yrs with S.D 12 yrs, what is the Mean and S.D of of Mobile users in a sample of 100 persons?

Sol: $\mu = 30, \sigma = 12, \quad n = 100$

By C.L.Th:

$$\text{Var}(\bar{x}) = \frac{\sigma^2}{n} = \frac{144}{100} = 1.44$$

$$SD(\bar{x}) = \frac{\sigma}{\sqrt{n}} = \frac{12}{\sqrt{100}} = 1.2$$

& Mean $(\bar{x}) \cong \mu = 30$ yrs.

i.e. $\bar{x} \sim N\{30, 1.44\}$

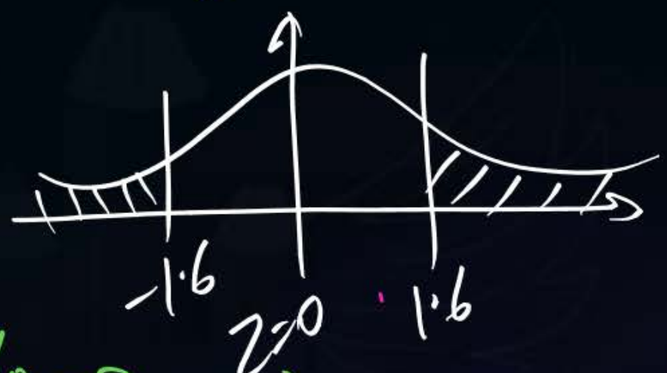
i.e. $\bar{x}$ is Normally Distributed by C.L.T

(ii) Also find the prob that Av. age of Mobile users in LKO city is less than 28 yrs if it is given that we have Confidence limit of 44.52% from $Z=0$ to 1.6

Sol: By Part (i), $\boxed{\bar{x} \sim N\{30, 1.44\}}$ i.e $\mu_{\bar{x}} = 30$, $\sigma_{\bar{x}} = 1.2$

$x = \{$ Av age of Mobile users in LKO city $\}$

$$Z = \frac{x - \mu_{\bar{x}}}{\sigma_{\bar{x}}} = \frac{28 - 30}{1.2} = -1.6$$



$$P(x<28) = P(Z<-1.6) = P(Z>1.6) = 0.5 - P(0<Z<1.6)$$

$$= 0.5 - 0.4452 = 0.0548 = \frac{5.4}{100} = 5.4\%$$

THANK - YOU